

Using SWAT-CUP to add Elevation Bands to Subbasin files

- 1- In the ArcSWAT project database (*project.mdb*) export the table “watershed” as an Excel file.
- 2- Open this file in Excel and delete all columns except:
Subbasin Area ElevMin ElevMax
- 3- In a new column calculate ($\text{diff} = \text{ElevMax} - \text{ElevMin}$) and sort on diff.
- 4- Delete all the rows (subbasins) where the difference is too small to consider elevation band, keeping in mind that temperature changes about -6 to -15 °C/km and precipitation between 1 to 1000 mm/km .
- 5- Five elevation bands are usually quite sufficient, use the following formula to calculate the mid-elevation of each band:
 $E = (\text{ElevMax} - \text{ElevMin}) / (5 * 2)$

and calculate mid point of each band in five columns as follows:

$$\begin{aligned} E1 &= \text{ElevMin} + 1 * E \\ E2 &= \text{ElevMin} + 3 * E \\ E3 &= \text{ElevMin} + 5 * E \\ E4 &= \text{ElevMin} + 7 * E \\ E5 &= \text{ElevMin} + 9 * E \end{aligned}$$

- 6- Sort on Subbasin and copy the columns
Subbasin E1 E2 E3 E4 E5
(with headings) into a notepad file called **elev_band.txt**
It should look like this:

Subbasin	E1	E2	E3	E4	E5
213	973.4	1268.2	1563	1857.8	2152.6
214	988.2	1322.6	1657	1991.4	2325.8
235	754	822	890	958	1026
.....					

- 7- Edit the file elev_band.def with the correct number of elevation bands and the final number of subbasins (rows) in step 6 above
- 8- Copy TopoRep.txt from your Arc_SWAT project (c:\....\ Watershed\text) into your working directory
9. The TopoRep.txt file has the annoying dashes! Run the program **replace.exe** to get rid of these dashes. This may get rid of a few minuses also for negative elevation bands, but this does not cause an important problem! The output of this program in **topo.txt**

10. Run the **Elev_band_CUP.exe** program. The program creates a file called model.in. This file contains the values of elevation as well as the area fraction for the desired subbasins:

```
v__ELEV_B(1).sub_____1      1583.800
v__ELEV_B_FR(1).sub_____1    0.043
v__ELEV_B(2).sub_____1      1951.400
v__ELEV_B_FR(2).sub_____1    0.692
v__ELEV_B(3).sub_____1      2319.000
v__ELEV_B_FR(3).sub_____1    0.200
v__ELEV_B(4).sub_____1      2686.600
v__ELEV_B_FR(4).sub_____1    0.055
v__ELEV_B(5).sub_____1      2686.600
v__ELEV_B_FR(5).sub_____1    0.001
.....
.....
```

11. Create a SWAT-CUP SUFI2 project and copy the file model.in into the swat-cup project directory.

12. Run the SWAT_Edit.exe program from the command line to replace the elevation bands and fractions into *.sub files.

13. Also, in SWAT-CUP set the TLAPS and PLAPS by creating another model.in file with the following content:

```
v__TLAPS.sub      -6 (or any desired number)
v__PLAPS.sub      10 (or any desired number)
```

Run SWAT_Edit again to sets TLAPS to -6 °C/km and PLAPS to 10 mm/km in all .sub files.

For questions and comments write to:

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